



F. M. 20

Bachelor of Computer Applications
St. Columba's College, Hazaribag
Semester-V, Internal Examination 2025
Design and Analysis of Algorithm[C5002]

Time :1.30hrs

Answer any 2 questions. Each question carry equal marks.

1. What is Asymptotic Notations? Differentiate between asymptotic notations Big Oh(O), Big Omega(Ω) and Big Theta(Θ)?
2. Find Time complexity of the following algorithms-

A. algo()

B. $f(n)=2n^2+3n+1$ show that $f(n)=O(n^3)$

Begin

```
Int l=1,s=1;
While(s<=n)
{
  l++;
  S=s+l;
  print "BCA-5"
}
```

End

3. What is Recursive algorithm? Define Basic steps and recursive step of a recurrence relation with example.
4. What is substitution method? Solve the following recurrence by using Substitution method
 $T(n)=n+T(n-1), n>1$

$=1$ $n=1$ OR

What is recursion tree method? Solve the following recurrence by using Recursion Tree Method

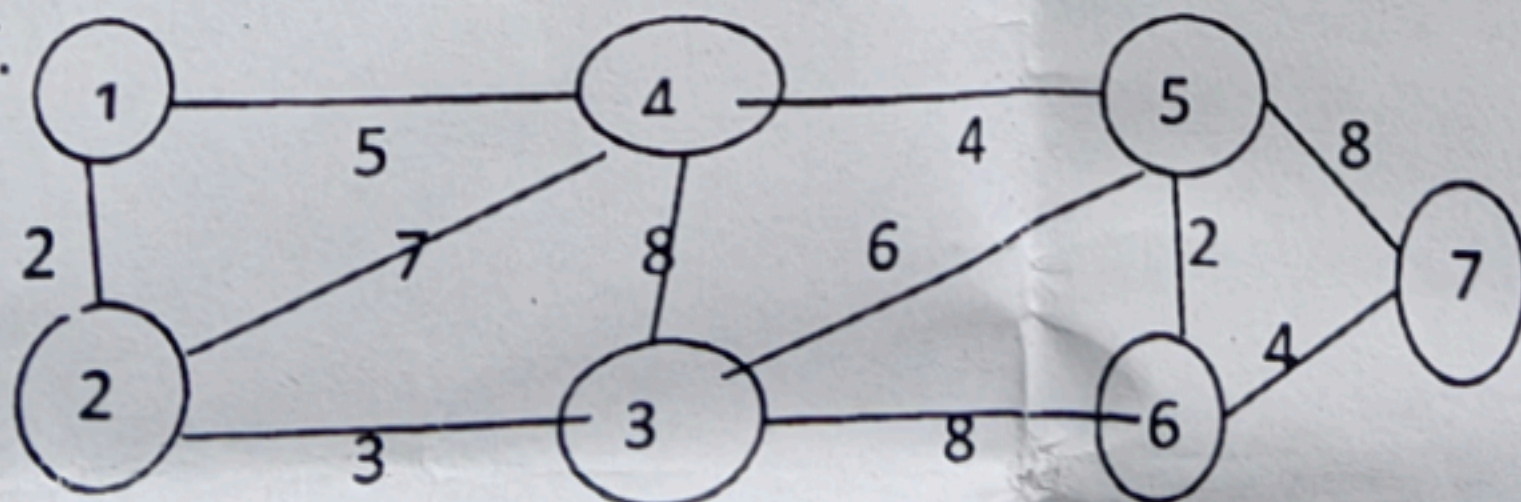
$T(n)=2T(n/2)+n, n>=1$

$=1$ $n=1$ OR

Explain Master Method. Solve the following recurrence by using Master Method

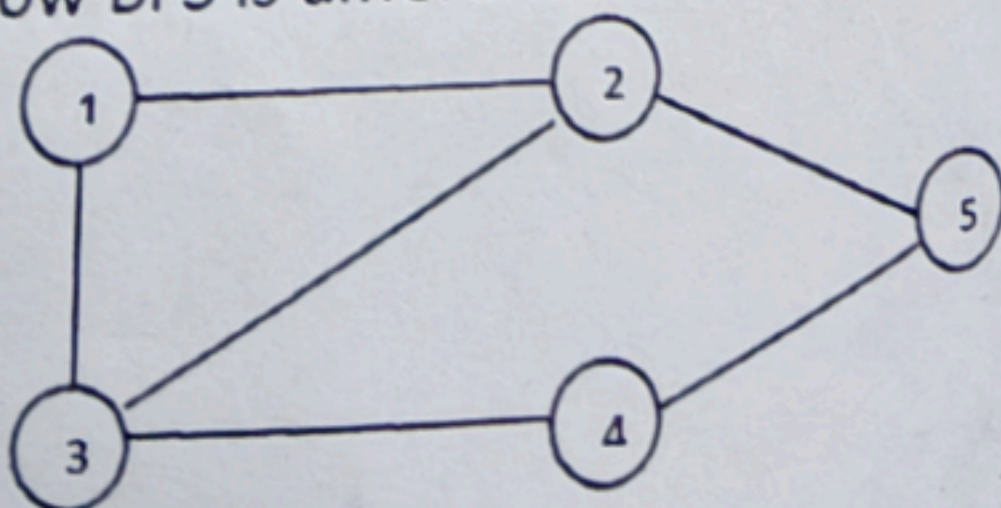
$T(n)=2T(n/2)+n/\log n$

5. Write Quick Sort algorithm by using divide and conquer techniques with time & space complexity of algorithm. OR
- A. Write Binary Search algorithm by using divide and conquer techniques with time complexity of algorithm.
- B. Write algorithm to find maximum and minimum in the given array by using divide and conquer techniques with time complexity of algorithm.
6. What is greedy technique? Write general form of greedy algorithm.
7. What is Minimum Cost Spanning Tree? Apply Kruskal's algorithm on the following graph to find MCST.



9. What is Hashing? Explain different types of Hash Function with proper example.

10. What is BFS? How BFS is different from DFS? Explore the vertices of the following graph by using BFS





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Bachelor of Computer Applications
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Semester-V, Internal Examination 2025
Computer Oriented Numerical Methods[C5004]

Time :1.30hrs

Answer any 4 questions. Each question carry equal marks.

1. Determine the roots of the equation $\cos x - x e^x = 0$, using Regula falsi method, correct to four places of decimal.
2. Compute i. $30^{-1/5}$ ii. $3\sqrt{24}$ iii. $18^{1/3}$, by Newton Raphson Method
3. Solve the following by Gauss-Jacobi Method.
 $20x+y-2z=17$, $3x+20y-z=-18$ and $2x-3y+20z=25$
4. From the following table, estimate the number of students who obtained marks between 40 and 50

Marks	30-40	40-50	50-60	60-70	70-80
No of students	31	42	51	35	31

5. Given the following

x	5	7	11	13	17
F(x)	150	392	1452	2366	5202

6. Solve q. n. 5 by using Newton's divided difference formula.
7. Establish the relations-

i. $\Delta = E \nabla = \delta E^{\frac{1}{2}} = \delta E^{-1/2}$ ii. $\mu = 1/2(E^{\frac{1}{2}} + E^{-1/2})$

evaluate f(x)
in table
larger form



Bachelor of Computer Applications

St. Columba's College, Hazaribag
Semester-V, Internal Examination, 2025
Linux Programming (C5003)

Time: 1:30 hrs

F. M. :20

Answer any 2 questions

1. What is Linux System? What are the silent features of Linux System?
2. What is file System in Linux? Explain various types of files available in Linux?
3. Define Shell and their types with their feature?
4. Explain Editors in Linux and their types? Explain different modes of Vi Editor?
5. What are the various Security provided by Linux system?
6. Explain the Commands with a suitable example?

- | | | |
|----------------|-----------|------------|
| a) Redirection | b) Piping | c) Filters |
| d) Grep | e) Sed | f) Pid |

or

- | | | |
|-----------|----------|------------|
| a) chkdsk | b) chmod | c) crontab |
| e) du | f) pwd | g) lpstat |

7. Write a Shell Program to enter any number and check that given number is Prime or Not?

Or

Write a shell program to generate Fibonacci Series up to given term?

Or

Write the syntax and an array program in Linux with suitable example?

8. What are the role and responsibility of DBA in Linux?

9. Sort Notes on (any 2)

- a) Metacharacters
- b) AWK
- c) Hostname
- d) System call
- e) Ipconfig

10. Write down the steps, how to compile and run C program in Linux Environment?

Write a C program to generate factorial of any given no?



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Bachelor of Computer Applications
St. Columba's College, Hazaribag
Semester-V, Internal Examination 2025
Internet Concept & Web Design [BCA C5001]

Time : 1 hrs

Each question carry 1 marks.

1. The common element which describe the web page, are ?
2. From which tag descriptive list starts ?
3. HTML uses User defined tags/ Pre-specified tags/Fixed tags defined by the language/ Tags only for linking
4. tag used to make a bulleted list with numbers?
5. HTML web pages can be read and rendered by
6. HTML stands for?
7. tag is used to mark a beginning of paragraph.
8. Correct HTML tag for the largest heading is
9. Correct HTML to left align the content inside a table cell is
10. What is the correct HTML for adding a background color?
- 11 - Which of the following is ~~not~~ a implicit object?
12. What JSP stands for ?
13. An array can be sorted in a reverse manner through method
14. Numeric array values can be sorted through an array method named as-
15. Attribute used in HTML for cell span of many columns.
16. If a computer in a network accesses resources that are shared by other computers it is called a
17. What does XML stand for?
18. Well formed XML document means
 - A. it contains a root element
 - B. it contain an element
 - C. it contains one or more elements
 - D. must contain one or more elements and root element must contain all other elements
19. What does DTD stand for?
20. Whats so great about XML?
 - A. Easy data exchange
 - B. High speed on network
 - C. Only B. is correct
 - D. Both A. & B.



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Bachelor of Computer Applications
St. Columba's College, Hazaribag
Semester-V, Internal Examination 2025
E-Commerce[C5007]

Time :1.30hrs

Answer any 2 questions. Each question carry equal marks.

1. What is e-commerce? Differentiate traditional commerce and e-commerce. (10)
2. What are ^{models}types of e-commerce? (10)
- 3 (a) What is World Wide Web? (5)
(b) What are software requirements for e-commerce? (5)
4. Write short notes on: (10)
(a) Digital signature (b) Firewalls
5. Explain the Security threats in e-commerce. (10)
6. What are payment methods used in e-commerce? (10)
7. Explain ethical, social and political issues in e-commerce. (10)